Universitat Pompeu Fabra Barcelona

MTG Music Technology Group

Music Technology Lab S103

An Interface for Generative Audio Latent Interpolation

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Course Structure

Class 1. Course Introduction Part 1 ✅
Class 2. Course Introduction Part 2 ✅
Class 3. Project Introduction Part 1: Background + Dataset Curation ⬅️
Class 4. Project Introduction Part 2: Training SCAPES + Project Details
Class 5 and so on. Project Development…

What is machine learning and what is it for?

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Machine Learning Basics

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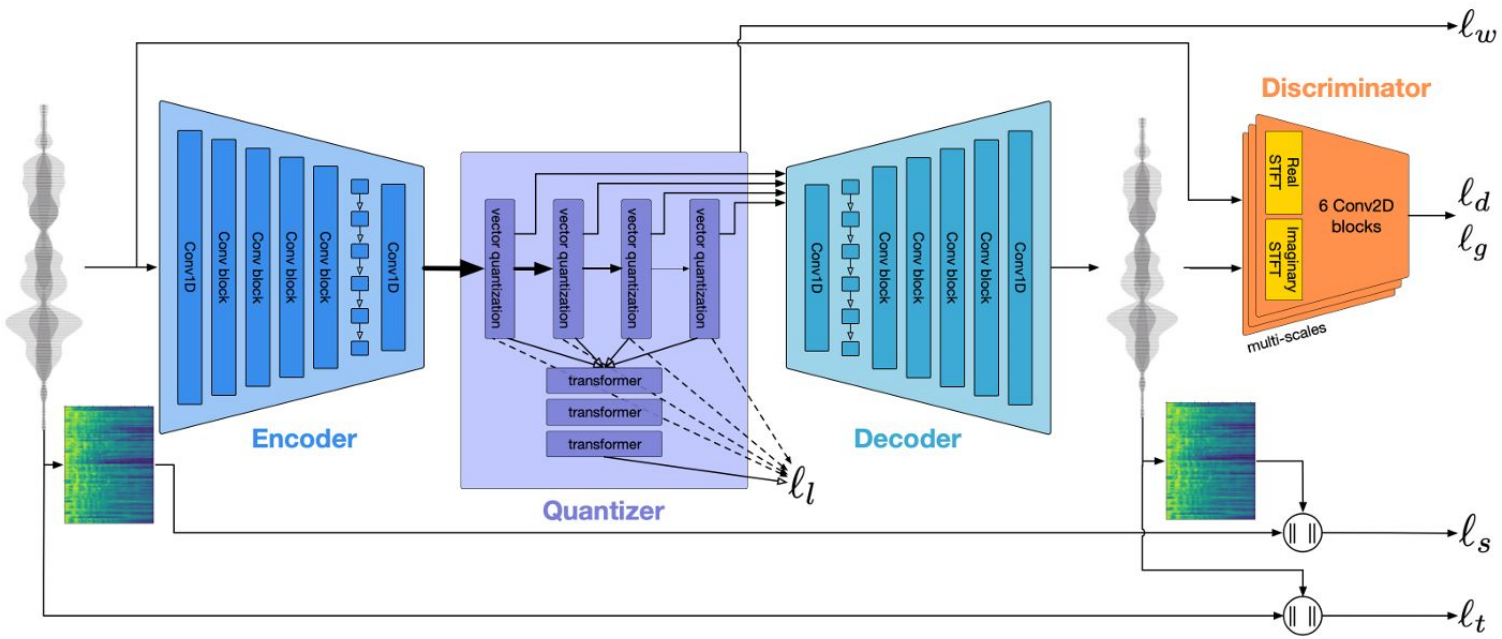
Training a model is the process of choosing its parameters based on data.

EnCodec

Meta's EnCodec is a neural audio codec — a machine-learning model that compresses audio into a compact latent representation and then reconstructs it with high fidelity. It's trained to encode raw audio into a lower-rate discrete form (via an encoder and quantizer) and then decode it back to sound, achieving much better compression than traditional codecs like MP3 while keeping quality high. This makes it useful for real-time streaming, bandwidth-efficient transmission, and as an intermediate audio representation for other models.



EnCodec

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Practical Stuff

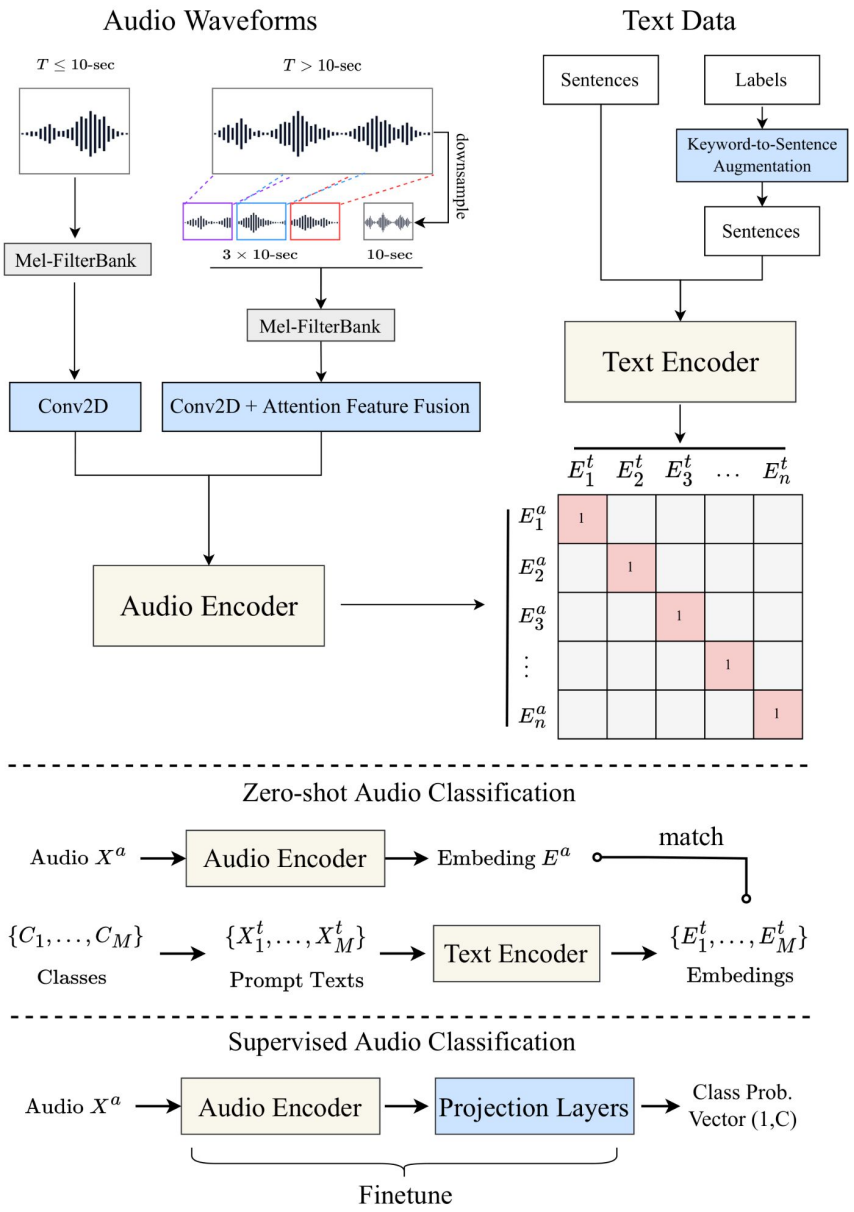
1. EnCodec representation is very detailed and universal – it can be used on any type of audio.
2. Different versions exist – main difference is sampling rate.
3. Trained on lots of data, including Freesound, so it knows many real-world sounds.
4. Easy to install in Python – we'll do a hands-on demo later.

What is it for?

1. Audio compression – reduce file size while keeping high-quality sound.
2. Streaming & real-time applications – lower bandwidth for live audio or cloud services.
3. Intermediate representation – as a latent audio embedding for other machine learning models.

CLAP: Contrastive Language-Audio Pretraining

CLAP (Contrastive Language-Audio Pretraining) is a machine learning **model** that **trains** on *pairs of audio and text descriptions* **data** so both modalities are mapped into a shared *common space* where related audio and language representations are close together, allowing the model to create audio representations that are semantically meaningful.



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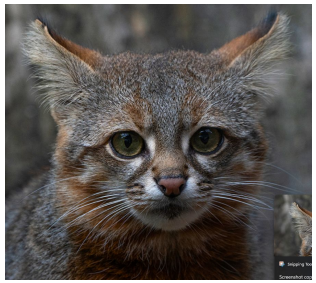


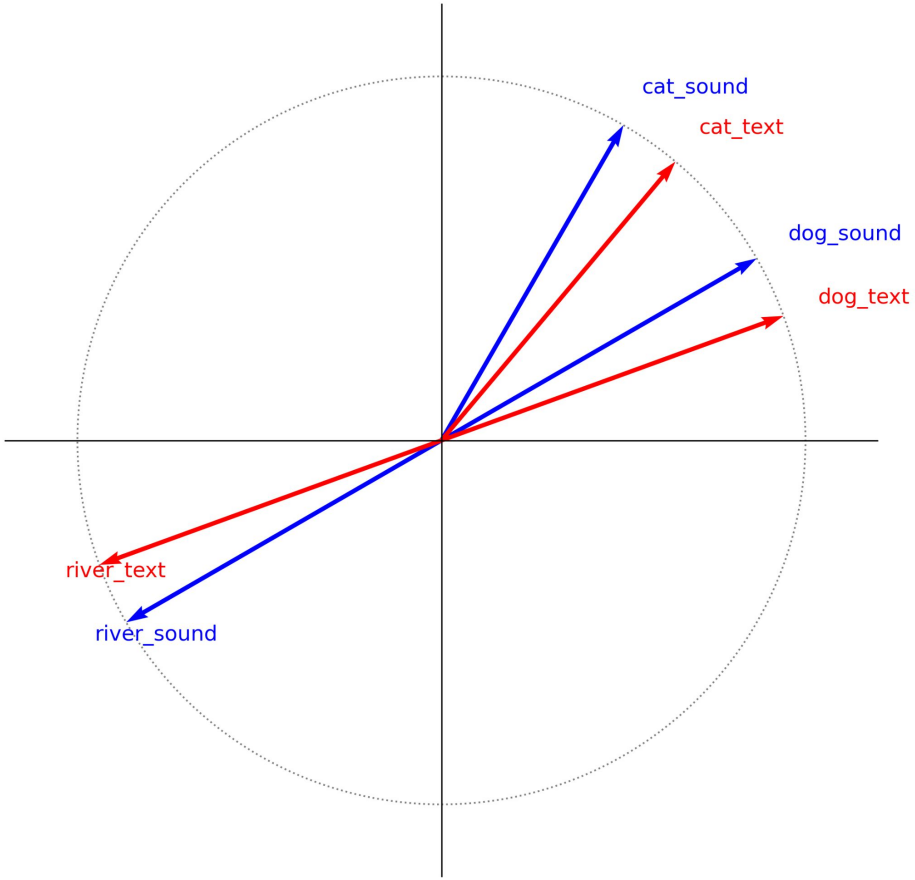
CLAP

Colo colo cat

Patagonian sheepdog

Palena river

CLAP



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What is it for?

1. Sound classification – recognize what type of sound is in a clip.
2. Audio search – find clips by text queries like "dog barking" or "rainstorm."
3. Cross-modal tasks – connect audio with language for tagging, recommendation, or retrieval.
4. Research & experiments – a starting point for building custom audio-language models.
5. Conditioning parameter – coarse feature that can be used to guide sound generation.

Dimensionality Reduction

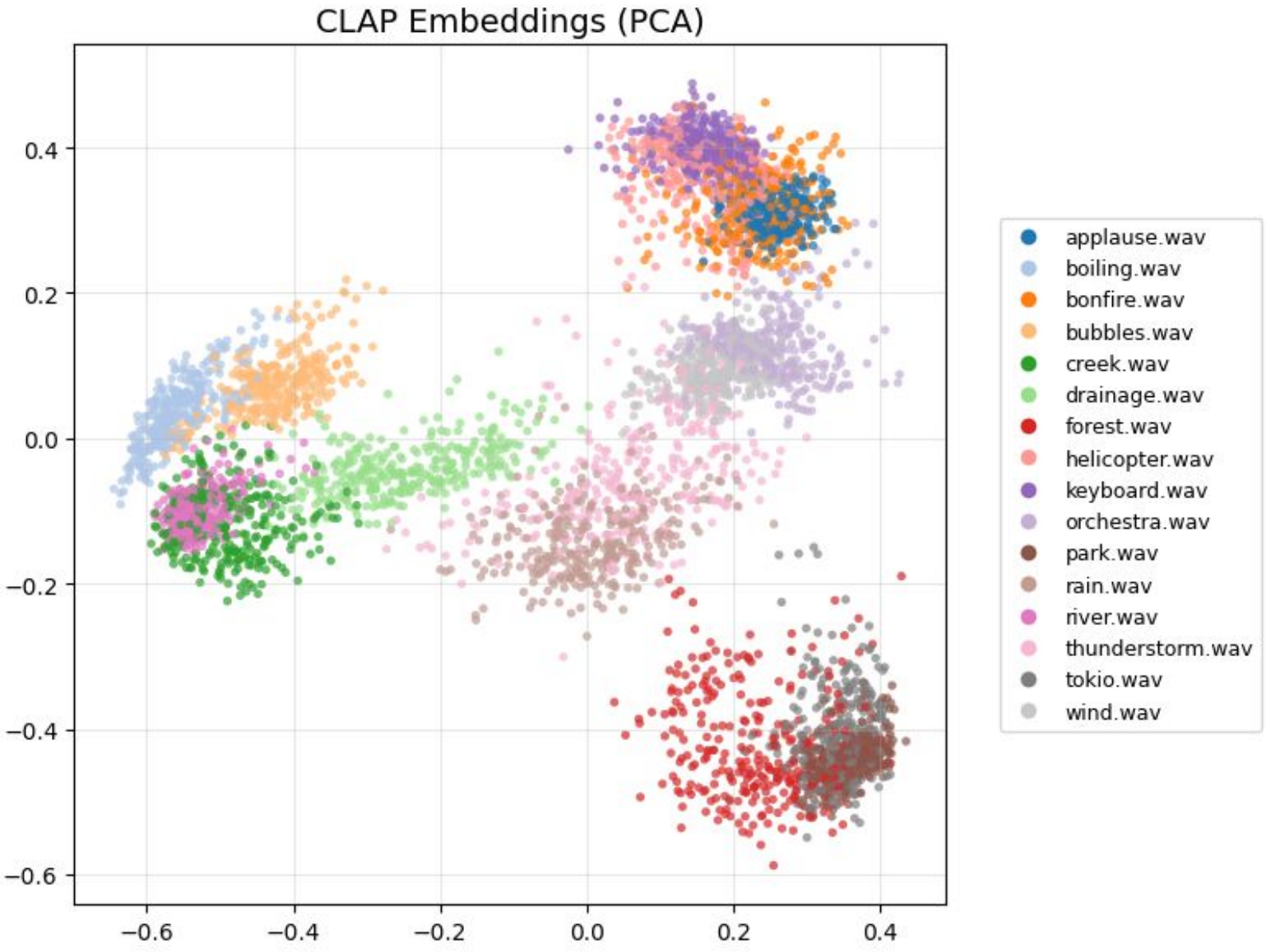
What it is:

Dimensionality reduction is a machine learning technique that transforms high-dimensional data into a lower-dimensional space while preserving important patterns and relationships. This makes it easier to visualize, understand, and process complex datasets.

Examples:

Common methods include **PCA (Principal Component Analysis)**, which finds directions of maximum variance; **t-SNE**, which preserves local relationships for visualization; and **UMAP**, which maintains both local and global structure efficiently for large datasets.

Dimensionality Reduction



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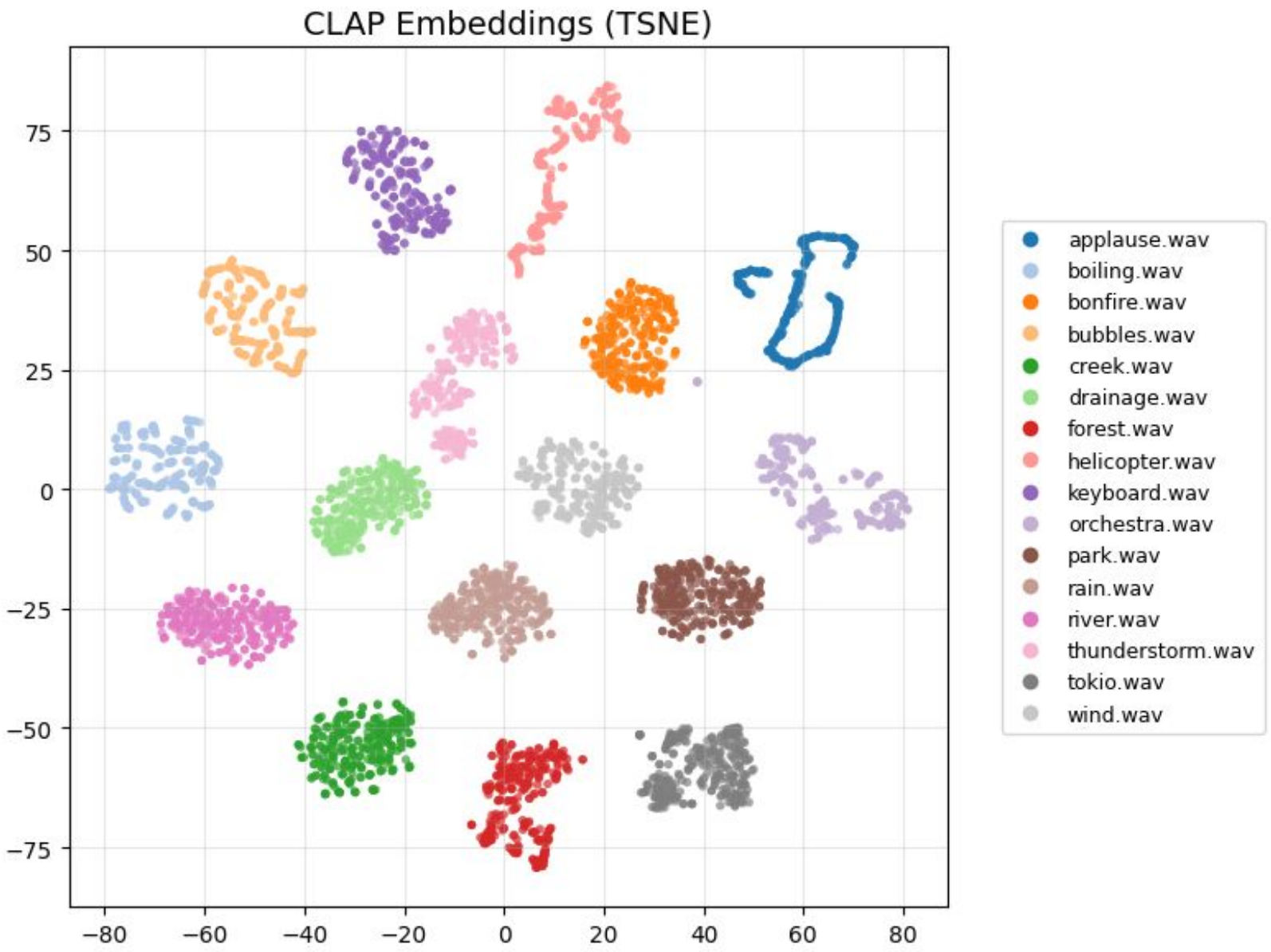

Thanks for listening!